

Prakruthi Nagaraj

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Full Stack Engineer

Software Engineer proficient in Python, C/C++, JavaScript, and ML, with a background in automation tools and real-world IoT projects. I'm a versatile and impact-driven engineer with experience across the full software development lifecycle, including RESTful APIs, responsive front-end development, scalable backend architectures, and AI-driven solutions. My mission is to build intelligent, reliable, and user-centered applications that harness the power of AI and modern web technologies to solve meaningful problems.

Technical Skills

Machine Learning | AI-driven Solutions | Python | C++ | JavaScript | HTML | CSS | SQL | Data Structures | Git | Docker | RESTful APIs | ELK Stack | Authentication | CI/CD | Testing | Version Control (Git) | Scalable Architectures

Technical Projects

Spots — HTML5 | CSS3 | JavaScript | REST API [GitHub](#) | [Live Demo](#)

Responsive Image-Sharing Web App for casual photographers to share and interact with scenic photos on all devices.

- Built user-facing features including profile editing, card creation, and like/unlike functionality.
- Implemented client-side validation and integrated RESTful APIs to enable persistent data operations with a remote server.
- Configured Webpack and optimized responsive layouts using media queries, enhancing performance across desktop and mobile platforms.
- Interpreted UI designs from Figma and deployed the app via GitHub Pages for easy public access.

Border Security Intrusion Detection — IoT | Face Recognition

Designed for military and border forces to automate intrusion detection and respond to real-time threats using smart surveillance.

- Built a face recognition module using Haar Cascade and image similarity search to identify intruders in live video feeds.
- Integrated remote controls using Blynk for electric fences, camera movement, and gate operation, enhancing field operability.
- Utilized Raspberry Pi and PIR sensors to trigger alerts and notify personnel instantly upon unauthorized movement.

Student Management System — Python | MySQL | REST API

A secure, role-based academic portal enabling administrators to manage students, courses, and staff accounts efficiently.

- Designed and developed RESTful APIs to manage CRUD operations for student, course, and user modules.
- Structured an ER diagram and built backend logic for multi-user roles and system interactions.
- Implemented hashed passwords and role-based access controls using Python and SQL, ensuring secure handling of sensitive data.

Professional Experience

Software Engineer, L&T Technology Services — Python | Automation | Jinja2

May 2021 – Nov 2022

- Developed a Python-based automation engine to replace legacy shell and Perl scripts, resulting in seamless PDK validation and layout generation across EDA tools.
- Reduced memory usage by ~60% and accelerated execution time of physical design flows (Layout Versus Schematic, Design Rule Check, and simulation logs) by optimizing data handling and eliminating redundancy.
- Utilized Jinja2 templating to dynamically generate validation and configuration files, enabling scalable automation across multiple semiconductor kits.
- Collaborated with cross-functional teams of design engineers, testers, and project leads to ensure delivery of robust, production-ready solutions aligned with project timelines.

Software Engineer Intern, L&T Technology Services — Python | ELK Stack | Telecom

Feb 2021 – Apr 2021

- Completed in-depth training on telecom protocols including 2G, 3G, 4G LTE, and 5G, preparing for real-world development in networking systems.
- Designed and developed a configurable log parser using Python to extract and structure data from raw telecom logs, enhancing debugging and system analysis workflows.
- Worked with ELK Stack (Elasticsearch, Logstash, Kibana) to visualize log data, enabling faster identification of telecom anomalies and improving reporting clarity.

Education

Software Engineering Certificate, TripleTen

2025

Machine Learning Specialization, Coursera

2025

BTech in Computer Science, REVA University, Bengaluru

Jun 2020